

REVIEW ARTICLE

# Large Language Models in Informed Consent — Opportunities, Evidence, and Challenges

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## Abstract

Informed consent forms (ICFs) are foundational to ethical clinical research consent, yet they remain lengthy, cognitively demanding, and written at reading levels that are not comprehensible to most participants. Recent advances in artificial intelligence, particularly large language models (LLMs), offer new opportunities to improve clarity, comprehension, accessibility, and participant engagement throughout the consent process. In this narrative review, the authors map the emerging landscape of LLM applications for ICFs across seven domains: glossary generation, plain language revisions, visual aid generation, accessibility and document formatting, translation into other languages, teach-back and comprehension validation, and interactive chat-based consent. The authors conclude by outlining considerations for their proper usage and integration with research and ethical review frameworks.

## Introduction

**I**nformed consent is central to the ethical conduct of clinical research, yet current consent materials and practices present substantial barriers to potential participants. The process includes providing participants with information about their rights; the study question and intervention; and the risks, benefits, and alternatives needed to support an informed decision. It extends from the initial trial invitation through to the trial's conclusion. Most informed consent forms (ICFs) span over 20 pages,<sup>1</sup> contain technical language that exceeds the recommended eighth-grade reading level, and contribute to documented gaps between perceived understanding and actual recall of key study elements.<sup>2,3</sup>

Large language models (LLMs) can be used to generate draft content for investigator review prior to submission to a research ethics committee (REC) (or institutional review board in the United States), after which the approved ICF is used unless subsequent revisions undergo rereview and approval. Alternatively, LLMs may be applied after REC approval; in these cases, the REC may not review the exact version presented to participants, raising concerns about content accuracy and fidelity. In addition, LLMs can produce plain language summaries, adapt structure to different literacy and accessibility needs, and support interactive exchanges that check and reinforce understanding.<sup>4-10</sup>

Here, we conducted a narrative review to examine the emerging role of LLMs in the clinical research informed consent process. We organize existing work into seven domains:

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| <b>Application Domain</b>    | <b>Potential Use Cases</b>                                                                                               | <b>Key Resources and Examples</b>                                                                                                                  | <b>Principal Risks</b>                                                                   | <b>Recommended Mitigation</b>                                                                                                                    |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Glossary generation          | Flags technical terms and links them to standardized plain language definitions                                          | MRCT Center Clinical Research Glossary <sup>11</sup> ; University of Florida IRB Glossary <sup>12</sup>                                            | Hallucinated or incomplete definitions; sponsor-biased framing                           | Constraint to preapproved sources; mandatory stakeholder review                                                                                  |
| Plain language rewriting     | Simplifies sentence structure and vocabulary to reduce reading level while preserving approved content                   | GPT-4 at Brown University Health (formerly Lifespan Healthcare) <sup>13</sup> ; multicenter GPT-4 study <sup>14</sup> ; Mixtral 8x22B <sup>5</sup> | Omitted qualifiers; understated risks; clinically significant omissions                  | Human review; constraint to source transformation; version control                                                                               |
| Visual aid generation        | Converts approved protocol elements into standardized visual representations of procedures, timelines, and risks         | RUNA icon system <sup>15</sup> ; video consent platforms <sup>16</sup>                                                                             | Omitted steps; misstated frequencies; overtrusted inaccurate images                      | Locked templates; derivation from REC-approved text; human review                                                                                |
| Accessibility                | Reformats consent materials across modalities to support diverse abilities, literacy levels, and assistive technologies  | VIC multimedia platform <sup>17</sup> ; RUNA icon-mapping system <sup>15</sup>                                                                     | Hallucinated benefits; mislabeled topics; privacy concerns                               | Retrieval-based constraints; privacy-compliant deployment; usability testing                                                                     |
| Translation                  | Produces multilingual consent materials or draft translations to support enrollment of non-English-speaking participants | GPT-4o Spanish <sup>18</sup> ; ChatGPT-3.5 multilingual <sup>19</sup> ; ChatGPT-4o comparison <sup>20</sup>                                        | Degraded performance for digitally underrepresented languages; hallucinated translations | Restriction to validated language pairs; mandatory human review for low-resource languages                                                       |
| Teach-back and comprehension | Generates summaries and comprehension checks to assess understanding                                                     | Gia chatbot platform <sup>21,22</sup> ; GPT-4 MCQA generator <sup>23</sup>                                                                         | Factual errors in questions; drift from approved language; scoring bias                  | Retrieval-augmented outputs; prereviewed templates; audit logs                                                                                   |
| Chatbot-assisted consent     | Delivers consent information conversationally, answers participant questions, and supports asynchronous engagement       | Gia platform <sup>21,22</sup> ; ChatGPT surgical documentation <sup>24</sup>                                                                       | Hallucinated content; overreliance on fluent but inaccurate responses                    | Retrieval-augmented responses; automated deviation checks; low-threshold escalation to the research team for clarification and consent questions |

\*GPT denotes Generative Pretrained Transformer; IRB, institutional review board; MCQA, multiple-choice question-answer pair; MRCT Center, Multi-Regional Clinical Trials Center; REC, research ethics committee; and VIC, Virtual Multimedia Interactive Informed Consent.

glossary generation, plain language rewriting, visual aid generation, accessibility and document formatting, translation, teach-back and comprehension validation, and interactive chat-based consent (Table 1).<sup>11-24</sup> For each domain, we synthesize empirical evidence, summarize strengths and failure modes, and highlight key gaps and potential mitigation strategies. We conclude by discussing the technical limitations, ethical considerations, and legal uncertainties associated with integrating LLMs into consent workflows.

## Methods

The application of LLMs to informed consent spans multiple disciplines — artificial intelligence (AI), bioethics, clinical trials methodology, regulatory science, and health

literacy — and involves heterogeneous study designs that are not amenable to formal meta-analysis. We therefore undertook a narrative review. We searched PubMed, Google Scholar, and the ACM Digital Library from January 2018 through December 2025 using combinations of the following terms: “large language model,” “LLM,” “ChatGPT,” “GPT-4,” “generative AI,” “natural language processing,” “informed consent,” “consent form,” “clinical trial,” “readability,” “health literacy,” “translation,” “teach-back,” “chatbot,” and “accessibility.” We included peer-reviewed original research, preprints with empirical data, and regulatory or ethical guidance documents published in English that evaluated or discussed LLM applications relevant to any stage of the informed consent process. We excluded commentaries without empirical content, studies focused exclusively on clinical decision support without a consent component, and non-English

publications. Reference lists of included articles were hand-searched to identify additional relevant studies. We acknowledge this approach may not capture all relevant literature, and we explicitly identify where evidence remains preliminary or limited to single-institution studies throughout the review.

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## Artificial Intelligence Applications and Opportunities

### GLOSSARY GENERATION

ICFs often rely on medical, procedural, and regulatory terminology that many participants find difficult to understand.<sup>25</sup> Prior studies suggest that 60%–70% of participants do not read or understand ICF content.<sup>17</sup> LLM-structured and/or integrated glossaries can clarify technical or medical terms that may impede understanding.

Early standardized glossaries have focused on creating plain language definitions for terms that commonly appear in clinical research documents, including ICFs.<sup>11,12</sup> Examples include the Multi-Regional Clinical Trials Center's Clinical Research Glossary, developed with patient advocates and refined through public review,<sup>11</sup> and the University of Florida Institutional Review Board's glossary of lay terms for consent forms,<sup>12</sup> which both provide plain language definitions for frequent trial terms (e.g., placebo, randomized, blinding, protocol). These resources are specifically designed for use in patient-facing materials, including consent forms and study explanations.<sup>11,12</sup>

Building on these standardized resources, LLMs can scan an ICF to flag terms that exceed a target reading level and link them to vetted plain language definitions from approved sources or local REC glossaries, thereby reducing the need for ad hoc editing. In electronic ICFs, definitions can be surfaced inline or via tooltips. When no definition exists, LLMs can propose draft entries for review and possible inclusion in local or multisite glossaries.

Key limitations of LLM-assisted glossary approaches include increased document length, the need for standardized definitions, and the risk that LLM-defined terms may oversimplify or omit important nuance. These risks can be mitigated by prioritizing preapproved glossary sources, clearly identifying newly proposed definitions, and treating all LLM-generated content as draft text subject to standard consent review and approval processes before it is shared with participants.

### PLAIN LANGUAGE REWRITING OF ICFs

Despite long-standing recommendations to write consent materials at or below an eighth-grade reading level, ICFs consistently exceed this threshold.<sup>26-29</sup> Recent analyses demonstrate that the problem persists: Coronavirus disease 2019 vaccine trial ICFs averaged an 11th-grade Flesch-Kincaid reading level,<sup>27</sup> and 798 ICFs of federally funded trials averaged a high school graduate reading level.<sup>29</sup> LLMs can systematically reduce sentence complexity; replace medical terminology with common equivalents; increase active voice usage; and restructure dense paragraphs into shorter, more digestible sections. LLMs can screen ICFs for sections that are above target readability, providing discrete steps to implement potential edits (e.g., concept explanation, terminology simplification, sentence restructuring, and formatting). LLMs can also produce parallel versions of an ICF that are tailored to accommodate participants with varying literacy skills.

Recent studies have shown that LLMs can improve readability while preserving accuracy and completeness.<sup>13,14,30</sup> For example, Generative Pretrained Transformer 4 (GPT-4) reduced Flesch-Kincaid grade reading levels from 12.6 to 6.7 for surgical consent at a major health care system.<sup>13</sup> Internationally, employing ChatGPT to recommend edits increased Gulpease Index scores for Italian vaccination consent forms from 36 to 82, indicating improved readability.<sup>30</sup>

However, simplified text may omit critical qualifiers or subtly alter risk descriptions, with studies documenting omission rates of 3%–4% in LLM-generated medical text.<sup>31</sup> This risk is addressable by completion of plain language rewriting prior to REC submission, allowing the investigator to correct the proposed final document. Accordingly, mitigation strategies should include tracked comparisons with the original ICF, clinician and research staff review of all edits and changes, and version control to preserve traceability.

### GENERATION OF VISUAL AIDS

Visual aids, such as icon arrays, pictographs, and videos, have been shown to improve comprehension of health information and risks, particularly among individuals with lower health literacy and/or numeracy.<sup>32-34</sup> LLMs can facilitate visual explanation by parsing protocol text and generating structured summaries of study-specific elements that can be translated into visual aids.

Several groups have already deployed AI or related technologies to generate visual consent aids and evaluated their impact.<sup>15,16</sup> RUNA, an interactive AI-based web

application, analyzes ICFs using machine learning models trained on 1623 German consent forms to identify 20 topics (e.g., risks, benefits, alternatives) and maps them to a standardized icon language.<sup>15</sup> In an initial user study of 19 ICF recipients, researchers reported a generally positive user experience, with participants noting that the icons were coherent, implied the underlying content, and aided in navigation.<sup>15</sup>

LLMs could extend these systems by generating trial-specific visual specifications directly from ICFs or protocols, reducing reliance on hand-crafted graphics. For example, an LLM could extract the visit schedule, randomization scheme, and potential major adverse events, and output structured instructions for an image pipeline to render visit timelines, randomization flow diagrams, and probability-based icon arrays for common and rare adverse events and side effects. With participant permission, visuals could also be personalized to better reflect the individual's experience. For example, generic avatars in study diagrams could be adapted to reflect self-identified characteristics such as family composition, hair color, or age, and visit timelines could be annotated with participant-relevant logistics such as a calendar, estimated travel time, childcare needs, and reminders. Personalization of this kind may increase engagement and help participants envision themselves in the study, but, without appropriate safeguards, it may also reinforce stereotypes or underrepresent some groups.

Without fidelity checks, visuals may omit steps, misstate frequencies, or invite undue reliance despite subtle inaccuracies. In many cases, visual aids can be generated in advance and reviewed by the study team and the REC before use, mitigating the risk of misinterpretation. Accordingly, visuals should be derived from a structured representation of the protocol and ICF, using locked templates for timelines and risk icon arrays, and maintaining version control for updates.

## ACCESSIBILITY

Over 1 billion people globally, including 240 million children, live with a disability.<sup>35-37</sup> While informed consent materials should be designed to accommodate a wide range of abilities and capacity levels,<sup>38</sup> ICFs remain dense and poorly formatted, with limited white space, small fonts, and minimal support for assistive technologies.<sup>39</sup> LLMs can automatically restructure and restyle ICFs across modalities to better meet accessibility requirements, including support for sensory and cognitive impairment, language and literacy barriers, and assistive technologies (e.g., screen readers or alternative input devices).

LLMs and related technologies are being actively implemented to address accessibility challenges by simplifying and transforming complex medical communication.<sup>15,17,40</sup> In one study, AI-generated summaries led approximately 80% of participants to report better understanding compared with the original consent document.<sup>40</sup> Complementary non-LLM tools include Virtual Multimedia Interactive Informed Consent (VIC), which leverages graphics, video, and text-to-speech capabilities to address literacy barriers,<sup>17</sup> and RUNA, which provides screen reader-compatible views and optical character recognition (OCR) extraction for mobile accessibility.<sup>15</sup>

Building on tools such as VIC and RUNA, LLMs can act as a generative layer that continuously adapts these interfaces. For example, VIC's scripted videos can be drafted by LLMs at multiple literacy levels and in multiple languages, while RUNA's topic icon mappings can be enriched with plain language labels and screen reader-friendly alternative text. LLMs can also structure OCR-extracted ICFs into standardized sections for existing multimedia workflows.

These extensions introduce specific risks, including incomplete restatement of risks and benefits, mislabeling of topics during restructuring, and accessibility failures (e.g., incorrect reading order for screen readers or missing alternative text). These risks can be reduced and potentially eliminated through retrieval-augmented models that constrain outputs to REC-approved content formats, automated checks for omissions and accessibility compliance, human review of any newly generated text before submission to the REC, and formal usability testing with participants from diverse backgrounds.

## TRANSLATION INTO OTHER LANGUAGES AND DIALECTS

Translating consent materials enables participation by individuals who do not read or speak the language of the original ICF. In the United States, almost 22% of residents 5 years of age and older speak a language other than English at home, about double the share from 4 decades ago, underscoring the importance of multilingual consent materials to support equitable research participation.<sup>41,42</sup> Traditional professional translation services, while accurate, are costly, labor-intensive, and often unavailable within the time frames required for clinical trial enrollment.<sup>43</sup> LLMs offer a mechanism to accelerate translation workflows if quality standards are maintained.

Recent evaluations have shown that LLMs can match the performance of professional translators for high-resource

language pairs (i.e., those with abundant training data), such as English–Spanish and English–French.<sup>18</sup> In a cross-sectional study, GPT-4o showed no significant difference from professional translators for Spanish pediatric patient instructions, with 52% of professional ratings preferring the LLM translations with fewer mistranslations.<sup>18</sup> However, performance varies substantially by language.<sup>19,20</sup> ChatGPT-3.5 outperformed Google Translate for Spanish (3.8% vs. 18.1% errors) but underperformed for Vietnamese (24.2% vs. 10.6% errors) and Russian (35.6% vs. 41.6% errors), with instances of content fabrication in Vietnamese and Russian that would make patient use of such translations without thorough prior review inadvisable.<sup>19</sup>

LLMs can support consent translation at scale; multilingual models such as Mixtral 8x22B can translate ICFs directly from English into languages such as Spanish, thereby improving access for underrepresented populations.<sup>5</sup> LLMs can also support machine-assisted translation workflows by generating draft translations that human translators refine, potentially shortening turnaround times and easing workload.<sup>43</sup>

However, translation performance varies by language: Accuracy declines for digitally underrepresented languages, and dialectal variation introduces additional errors.<sup>43</sup> Because consent translation requires precise medical and legal equivalence, LLMs may produce plausible but incorrect, incomplete, or culturally inappropriate phrasing.<sup>19,43</sup> While these gaps may narrow as systems accumulate in-domain examples for iterative evaluation and fine-tuning,<sup>43</sup> initial deployments should be limited to validated language pairs, require human review for low-resource languages, and use consent-specific terminology glossaries.

### TEACH-BACK AND COMPREHENSION ASSESSMENT

Teach-back, a method that involves asking individuals to recall or explain, in their own words, what has been discussed, is recommended to confirm understanding.<sup>44</sup> A key barrier is time: The “teach to goal” approach requires repeated clarification until understanding is achieved and is widely perceived as adding substantial burden in both clinical and research settings.<sup>45</sup> LLMs can support teach-back by generating patient-friendly summaries and comprehension checks (e.g., multiple-choice question–answer pairs [MCQAs]) for complex documents, providing targeted clarification when participants answer incorrectly, and escalating to study staff when understanding remains insufficient.

Early deployments show that teach-back can be implemented at scale using automated agents.<sup>21,23</sup> In genomic

research, the chat-based consent platform Gia generates an REC-approved scripted conversation and a 10-item quiz to assess understanding.<sup>21</sup> In a retrospective analysis of 144 referred families, Gia users had shorter consent interactions than patients who participated in staff-led consent (median 44 minutes vs. 76 minutes) and completed consent sooner after referral (median 5 days vs. 16 days), with 86% of users rating the experience positively.<sup>21</sup> Beyond scripted systems, GPT-4 has been used to generate MCQAs directly from cancer trial ICFs; the questions showed high concordance with human-annotated answers, and more than 80% of surveyed patients reported improved understanding of a complex radiation trial after reviewing the questions.<sup>23</sup>

LLMs could extend these early systems in several concrete ways, such as by automatically generating section-specific questions from an REC-approved ICF, transforming free-text teach-back responses into structured scores and misconception flags, and powering chat companions that provide real-time clarification anchored to the consent document and related educational materials.

Key risks of LLM-assisted teach-back include factual errors or omissions, drift from REC-approved language with continued use over time, biased or inaccurate scoring (including for participants with limited English proficiency), and proceeding with enrollment despite inadequate understanding. These risks can be mitigated by constraining outputs to retrieval-augmented responses grounded in approved materials, using prereviewed questions and scoring templates that allow partial responses, and defining clear escalation pathways to study staff when understanding cannot be reliably assessed.

### CHATBOT-ASSISTED CONSENT

Not all clinical research depends upon staff-mediated consent; in minimal-risk survey studies, for example, consent is often implied, with survey completion and return serving as implicit agreement to participate. In most interventional clinical trials and genomics studies, however, consent relies on staff-mediated, time-intensive discussion and documentation, which can become a bottleneck and limit enrollment capacity at scale.<sup>22</sup> A chatbot-assisted process shifts consent from REC-approved invariant documents that are — or can be — subject to human review to an interactive, conversational engagement. LLM-powered chatbots can deliver consent information conversationally, answer questions in real time, and assess comprehension through integrated quizzes, while allowing participants to complete the consent process asynchronously at their own pace.



Early implementations of chat-based consent demonstrate feasibility, with outcomes comparable to those of traditional approaches.<sup>21,22</sup> As noted above, the Gia platform reduced time to consent by approximately two thirds with high participant satisfaction. A larger subsequent analysis replicated these findings, with consent often completed outside working hours, highlighting the need for flexibility and asynchronous completion.<sup>21</sup>

Unlike scripted, rule-based chatbots with fixed responses and deterministic branching, generative LLMs can answer novel participant questions by drawing on REC-approved consent documents and related educational materials. They can tailor explanations in real time, expanding when comprehension is incomplete and proceeding when understanding is confirmed. LLM chatbots can be continuously available, enabling engagement outside working hours and reducing scheduling barriers that may limit participation.<sup>46</sup>

However, generative LLMs introduce distinct risks. They may hallucinate content absent from REC-approved documents, and their conversational fluency can create an illusion of accuracy, leading participants to trust subtly inaccurate responses.<sup>24,47</sup> Mitigation should include constraining outputs to retrieval-augmented responses drawn exclusively from REC-approved content, implementing automated deviation checks, maintaining clear pathways for study staff to address unresolved or study-specific questions, and transparently communicating that participants are interacting with an AI-enabled tool.<sup>46</sup>

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## Discussion

This narrative review synthesizes the emerging landscape of LLM applications across seven domains of the clinical research informed consent process. The available evidence suggests that LLMs can meaningfully improve the readability and comprehension of consent materials and the efficiency of the consent process. However, substantial safeguards and multisite validation are needed before widespread deployment.

LLMs consistently demonstrate the capacity to reduce the linguistic complexity of consent materials. Plain language rewriting of study documents achieved reductions in Flesch-Kincaid grade levels from the 12th–14th grade range to the 7th–9th grade range, approaching the long-recommended 8th-grade threshold.<sup>13,14,26–30</sup> The Gia platform demonstrates that chat-based consent can reduce time to consent while maintaining 96% pass rates on comprehension quizzes.<sup>21,22</sup> Notably, participants frequently complete

chatbot-assisted consent outside traditional working hours, suggesting an underappreciated equity dimension: Asynchronous consent may reduce barriers for individuals whose responsibilities preclude daytime appointments.<sup>21,46</sup>

Despite these promising findings, the current evidence base remains limited. Most studies are derived from single institutions, use readability metrics rather than validated comprehension assessments, and focus on English-language documents. The translation literature illustrates how dramatically performance varies: Spanish translations approach professional quality, while Vietnamese and Russian translations exhibit high error rates with instances of fabrication.<sup>18–20</sup> This heterogeneity suggests LLM capabilities cannot be assumed to generalize without domain-specific validation. Future research should prioritize comparative trials that measure comprehension outcomes stratified by health literacy and language preference, and include content validation, hallucination and deviation detection systems, and participant-focused preferences with a particular focus on vulnerable and underrepresented populations.<sup>31,35–38,41,42</sup>

Moreover, the use of LLMs in ICFs introduces several technical, ethical, and legal concerns. First, unconstrained generation can introduce hallucinations and omissions; in medical applications, reported hallucination rates of around 1.5% and omission rates of 3%–4% would be unacceptable at scale.<sup>31</sup> Notably, no published standard defines an acceptable hallucination or omission rate for clinical AI applications, and the question of whether any nonzero error rate is tolerable in the consent context remains unresolved. Empirical evidence suggests that clinicians hold AI to a more stringent accuracy threshold than they apply to human performance,<sup>48</sup> and the ethical requirement for accurate risk disclosure in informed consent may demand an effectively zero-tolerance standard for fabricated content. While retrieval-augmented generation systems grounded in curated source documents have demonstrated substantial reductions in hallucination, in some constrained clinical domains achieving zero hallucination rates,<sup>49,50</sup> these systems reduce rather than eliminate risk, as they introduce a trade-off: Systems that decline to answer when source information is lacking may fail to address legitimate participant questions, while systems that generate beyond their source material reintroduce hallucination risk.<sup>49</sup>

Second, structural bias in LLM training data poses a direct threat to equitable consent. A systematic review of demographic disparities in medical LLMs found that over 90% of studies identified biases, with gender bias present in

93.7% and racial or ethnic bias present in 90.9% of studies that examined these biases.<sup>51</sup> For consent applications, these findings mean that LLM-generated materials could vary in content, tone, or completeness across demographic groups, undermining the ethical requirement for autonomy in informed consent. Safeguards should include systematic bias auditing across demographic groups, validation with diverse participant populations before deployment, and explicit testing across languages and dialects served by a given trial site.

Third, the liability implications of LLM-generated misstatements in consent materials remain unresolved. Under current tort frameworks, physicians and investigators who rely on AI-generated content bear primary malpractice liability, while LLM developers are largely shielded because AI software often does not qualify as a “product” under existing doctrine, and causation from algorithmic errors is difficult to prove.<sup>52,53</sup> No court has yet adjudicated liability for a medical injury arising from reliance on generative AI in the consent context.<sup>54</sup> For dynamic conversational consent, where LLM responses cannot be fully prereviewed, institutions should establish documentation and audit standards that include logging all participant-facing outputs, maintaining version-controlled records of the underlying model and retrieval corpus, conducting periodic automated and human review of interaction transcripts, and defining clear institutional responsibility for AI-generated content.<sup>52</sup> These governance measures are particularly important because adaptive consent interactions do not cleanly map onto current REC review paradigms designed for static documents.

Current implementations have operated within standard REC review mechanisms, suggesting that pathways need not change significantly when LLM outputs are static, reviewable, and version-controlled. However, dynamic participant-facing interactions raise new regulatory questions, as adaptive, generative responses cannot be reviewed and approved in advance by RECs. In this context, the process of informed consent, involving discussion and exchange between participant and study staff, is also adaptive, in the sense that it changes in response to participants’ questions and understanding, and is not approved, scripted, or generally monitored by RECs. Hybrid approaches combining scripted REC-approved content with structured comprehension assessments represent pragmatic near-term pathways.<sup>21,22</sup> Controls should be proportional to protocol risk and participant vulnerability, with more conservative designs for higher-risk studies. Given the pace of model development, establishing reliable evaluation, monitoring, and change control is essential for broad implementation.

LLMs show early but promising potential to address longstanding barriers in informed consent,<sup>13,14,18,21-23,30,43-45</sup> yet the same capabilities that make them valuable create risks where accuracy is paramount.<sup>24,31,47</sup> Developed thoughtfully, LLM-assisted consent should help realize the ethical ideal of demonstrated comprehension while expanding research access for populations challenged by literacy, language, or logistical barriers.<sup>41,42</sup>

## Disclosures

Author disclosures are available at [ai.nejm.org](https://ai.nejm.org).

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## References

1. King-Kallimanis BL, Ferris A, Dropkin L, et al. Initial steps in creating a patient-centric addendum to clinical trial informed consent forms. *JTO Clin Res Rep* 2023;4:100575. DOI: [10.1016/j.jtocrr.2023.100575](https://doi.org/10.1016/j.jtocrr.2023.100575).
2. Gourlay E, Felton T, Bafadhel M, et al. Readability and complexity of written information presented to hospitalised patients for trial consent during the COVID-19 pandemic in the UK: a retrospective document analysis. *BMJ Open* 2025;15:e089447. DOI: [10.1136/bmjopen-2024-089447](https://doi.org/10.1136/bmjopen-2024-089447).
3. Christine B, Daniel W, Florian L, et al. “Uninformed consent” in clinical trials with cancer patients: a qualitative analysis of patients’ and support persons’ communication experiences and needs. *Patient Educ Couns* 2024;122:108144. DOI: [10.1016/j.pec.2024.108144](https://doi.org/10.1016/j.pec.2024.108144).
4. Gelinas L, Morrell W, Tse T, Glazier A, Zarin DA, Bierer BE. Characterization of key information sections in informed consent forms posted on ClinicalTrials.gov. *J Clin Transl Sci* 2023;7:e185. DOI: [10.1017/cts.2023.605](https://doi.org/10.1017/cts.2023.605).
5. Shi Q, Luzuriaga K, Allison JJ, et al. Transforming informed consent generation using large language models: mixed methods study. *JMIR Med Inform* 2025;13:e68139. DOI: [10.2196/68139](https://doi.org/10.2196/68139).
6. Petrou E, Ormond KE, Stammach D, Ash E, Buchholz O, Vayena E. Evaluating GPT-4’s ability to generate informed consent material for genetic testing. *NPJ Artif Intell* 2025;1:32. DOI: [10.1038/s44387-025-00036-4](https://doi.org/10.1038/s44387-025-00036-4).
7. Van Driel MHE, Blok N, Van den Brand JA JG, et al. Leveraging GPT-4 enables patient comprehension of radiology reports. *Eur J Radiol* 2025;187:112111. DOI: [10.1016/j.ejrad.2025.112111](https://doi.org/10.1016/j.ejrad.2025.112111).

8. Cork SC, Hopcroft K. Evaluating ChatGPT for converting clinic letters into patient-friendly language: a quantitative study. *BJGP Open* 2025;9:BJGPO.2024.0300. DOI: [10.3399/BJGPO.2024.0300](https://doi.org/10.3399/BJGPO.2024.0300).
9. Currie G, Robbie S, Tually P. ChatGPT and patient information in nuclear medicine: GPT-3.5 versus GPT-4. *J Nucl Med Technol* 2023;51:307-313. DOI: [10.2967/jnmt.123.266151](https://doi.org/10.2967/jnmt.123.266151).
10. McMinn D, Grant T, DeFord-Watts L, et al. Using artificial intelligence to expedite and enhance plain language summary abstract writing of scientific content. *JAMIA Open* 2025;8:ooaf023. DOI: [10.1093/jamiaopen/ooaf023](https://doi.org/10.1093/jamiaopen/ooaf023).
11. Clinical Research Glossary. Multi-Regional Clinical Trials Center of Brigham and Women's Hospital and Harvard. (<https://mrctcenter.org/glossary/>).
12. University of Florida Institutional Review Board. Glossary of lay terms for use in informed consent forms. 2026 (<https://irb.ufl.edu/irb01/forms/glossary.html>).
13. Mirza FN, Tang OY, Connolly ID, et al. Using ChatGPT to facilitate truly informed medical consent. *NEJM AI* 2024;1(2). DOI: [10.1056/AIcs2300145](https://doi.org/10.1056/AIcs2300145).
14. Ali R, Connolly ID, Tang OY, et al. Bridging the literacy gap for surgical consents: an AI-human expert collaborative approach. *NPJ Digit Med* 2024;7:63. DOI: [10.1038/s41746-024-01039-2](https://doi.org/10.1038/s41746-024-01039-2).
15. Kruger AS, Moebert T, Muller I, Lucke U. An interactive AI-based web application for preparing informed consent forms comprehensibly to laypersons. In: *INFORMATIK 2025*. Potsdam, Germany: Gesellschaft für Informatik, 2025:867. DOI: [10.18420/INF2025\\_75](https://doi.org/10.18420/INF2025_75).
16. Abujarad F, Peduzzi P, Mun S, et al. Comparing a multimedia digital informed consent tool with traditional paper-based methods: randomized controlled trial. *JMIR Form Res* 2021;5:e20458. DOI: [10.2196/20458](https://doi.org/10.2196/20458).
17. Abujarad F, Alfano S, Bright TJ, et al. Building an informed consent tool starting with the patient: the patient-centered virtual multimedia interactive informed consent (VIC). *AMIA Annu Symp Proc* 2018;2017:374-383.
18. Ray M, Kats DJ, Moorkens J, et al. Evaluating a large language model in translating patient instructions to Spanish using a standardized framework. *JAMA Pediatr* 2025;179:1026. DOI: [10.1001/jamapediatrics.2025.1729](https://doi.org/10.1001/jamapediatrics.2025.1729).
19. Rao P, McGee LM, Seidman CA. A comparative assessment of ChatGPT versus Google Translate for the translation of patient instructions. *J Med Artif Intell* 2024;7:11. DOI: [10.21037/jmai-24-24](https://doi.org/10.21037/jmai-24-24).
20. Brewster RCL, Gonzalez P, Khazanchi R, et al. Performance of ChatGPT and Google Translate for pediatric discharge instruction translation. *Pediatrics* 2024;154:e2023065573. DOI: [10.1542/peds.2023-065573](https://doi.org/10.1542/peds.2023-065573).
21. Savage SK, LoTempio J, Smith ED, et al. Using a chat-based informed consent tool in large-scale genomic research. *J Am Med Inform Assoc* 2024;31:472-478. DOI: [10.1093/jamia/ocad181](https://doi.org/10.1093/jamia/ocad181).
22. Smith ED, Savage SK, Andrew EH, et al. Development and implementation of novel chatbot-based genomic research consent. January 24, 2023 (<https://www.biorxiv.org/content/10.1101/2023.01.23.525221v1>). Preprint.
23. Gao M, Varshney A, Chen S, et al. The use of large language models to enhance cancer clinical trial educational materials. *JNCI Cancer Spectr* 2025;9:pkaf021. DOI: [10.1093/jncics/pkaf021](https://doi.org/10.1093/jncics/pkaf021).
24. Decker H, Trang K, Ramirez J, et al. Large language model-based chatbot versus surgeon-generated informed consent documentation for common procedures. *JAMA Netw Open* 2023;6:e2336997. DOI: [10.1001/jamanetworkopen.2023.36997](https://doi.org/10.1001/jamanetworkopen.2023.36997).
25. Falagas ME, Korbila IP, Giannopoulou KP, Kondilis BK, Peppas G. Informed consent: how much and what do patients understand? *Am J Surg* 2009;198:420-435. DOI: [10.1016/j.amjsurg.2009.02.010](https://doi.org/10.1016/j.amjsurg.2009.02.010).
26. Paasche-Orlow MK, Taylor HA, Brancati FL. Readability standards for informed-consent forms as compared with actual readability. *N Engl J Med* 2003;348:721-726. DOI: [10.1056/NEJMsa021212](https://doi.org/10.1056/NEJMsa021212).
27. Bothun LS, Feeder SE, Poland GA. Readability of participant informed consent forms and informational documents. *Mayo Clin Proc* 2021;96:2095-2101. DOI: [10.1016/j.mayocp.2021.05.025](https://doi.org/10.1016/j.mayocp.2021.05.025).
28. Grossman SA, Piantadosi S, Covahey C. Are informed consent forms that describe clinical oncology research protocols readable by most patients and their families? *J Clin Oncol* 1994;12:2211-2215. DOI: [10.1200/JCO.1994.12.10.2211](https://doi.org/10.1200/JCO.1994.12.10.2211).
29. Mirza FN, Wu E, Abdulrazeq HF, et al. The literacy barrier in clinical trial consents: a retrospective analysis. *eClinicalMedicine* 2024;75:102814. DOI: [10.1016/j.eclinm.2024.102814](https://doi.org/10.1016/j.eclinm.2024.102814).
30. Cosma C, Radi A, Cattano R, et al. Potential role of ChatGPT in simplifying and improving informed consent forms for vaccination: a pilot study conducted in Italy. *BMJ Health Care Inform* 2025;32:e101248. DOI: [10.1136/bmjhci-2024-101248](https://doi.org/10.1136/bmjhci-2024-101248).
31. Asgari E, Montaña-Brown N, Dubois M, et al. A framework to assess clinical safety and hallucination rates of LLMs for medical text summarisation. *NPJ Digit Med* 2025;8:274. DOI: [10.1038/s41746-025-01670-7](https://doi.org/10.1038/s41746-025-01670-7).
32. Galmarini E, Marciano L, Schulz PJ. The effectiveness of visual-based interventions on health literacy in health care: a systematic review and meta-analysis. *BMC Health Serv Res* 2024;24:718. DOI: [10.1186/s12913-024-11138-1](https://doi.org/10.1186/s12913-024-11138-1).
33. Rosenfeld EH, Lopez ME, Yu YR, et al. Use of standardized visual aids improves informed consent for appendectomy in children: a randomized control trial. *Am J Surg* 2018;216:730-735. DOI: [10.1016/j.amjsurg.2018.07.032](https://doi.org/10.1016/j.amjsurg.2018.07.032).
34. Ancker JS, Senathirajah Y, Kukafka R, Starren JB. Design features of graphs in health risk communication: a systematic review. *J Am Med Inform Assoc* 2006;13:608-618. DOI: [10.1197/jamia.M2115](https://doi.org/10.1197/jamia.M2115).
35. World Health Organization. Disability. March 7, 2023 (<https://www.who.int/news-room/fact-sheets/detail/disability-and-health>).



36. United Nations Children's Fund. Seen, counted, included: using data to shed light on the well-being of children with disabilities. December 2024 (<https://data.unicef.org/resources/children-with-disabilities-report-2021/>).
37. Centers for Disease Control and Prevention. Disability impacts all of us infographic. April 14, 2025 (<https://www.cdc.gov/disability-and-health/articles-documents/disability-impacts-all-of-us-infographic.html>).
38. O'Hare F, Thrimawithana S, Clague A, et al. PROTOCOL: strategies to enhance inclusion in informed consent practice for people with vision and/or hearing support needs: a systematic review. *Campbell Syst Rev* 2025;21:e70065. DOI: [10.1002/cl2.70065](https://doi.org/10.1002/cl2.70065).
39. Level Access. The role of assistive technology in digital inclusion. September 9, 2024 (<https://www.levelaccess.com/blog/assistive-technology/>).
40. Waters M. AI meets informed consent: a new era for clinical trial communication. *JNCI Cancer Spectr* 2025;9:pkaf028. DOI: [10.1093/jncics/pkaf028](https://doi.org/10.1093/jncics/pkaf028).
41. United States Census Bureau. New data on detailed languages spoken at home and the ability to speak English. June 3, 2025 (<https://www.census.gov/newsroom/press-releases/2025/2017-2021-acs-language-use-tables.html>).
42. Muthukumar AV, Morrell W, Bierer BE. Evaluating the frequency of English language requirements in clinical trial eligibility criteria: a systematic analysis using ClinicalTrials.gov. *PLoS Med* 2021;18:e1003758. DOI: [10.1371/journal.pmed.1003758](https://doi.org/10.1371/journal.pmed.1003758).
43. Lopez I, Velasquez DE, Chen JH, Rodriguez JA. Operationalizing machine-assisted translation in healthcare. *NPJ Digit Med* 2025;8:584. DOI: [10.1038/s41746-025-01944-0](https://doi.org/10.1038/s41746-025-01944-0).
44. Kripalani S, Bengtzen R, Henderson LE, Jacobson TA. Clinical research in low-literacy populations: using teach-back to assess comprehension of informed consent and privacy information. *IRB* 2008;30:13-19.
45. Jager AJ, Wynia MK. Who gets a teach-back? Patient-reported incidence of experiencing a teach-back. *J Health Commun* 2012;17:294-302. DOI: [10.1080/10810730.2012.712624](https://doi.org/10.1080/10810730.2012.712624).
46. Rothstein MA. Should chatbots be used to obtain informed consent for research? *Ethics Hum Res* 2023;45:46-50. DOI: [10.1002/eahr.500190](https://doi.org/10.1002/eahr.500190).
47. Ghim J-L, Ahn S. Transforming clinical trials: the emerging roles of large language models. *Transl Clin Pharmacol* 2023;31:131. DOI: [10.12793/tcp.2023.31.e16](https://doi.org/10.12793/tcp.2023.31.e16).
48. Lenskjold A, Nybing JU, Trampedach C, et al. Should artificial intelligence have lower acceptable error rates than humans? *BJR Open* 2023;5:20220053. DOI: [10.1259/bjro.20220053](https://doi.org/10.1259/bjro.20220053).
49. Nishisako S, Higashi T, Wakao F. Reducing hallucinations and trade-offs in responses in generative AI chatbots for cancer information: development and evaluation study. *JMIR Cancer* 2025;11:e70176. DOI: [10.2196/70176](https://doi.org/10.2196/70176).
50. Wada A, Tanaka Y, Nishizawa M, et al. Retrieval-augmented generation elevates local LLM quality in radiology contrast media consultation. *NPJ Digit Med* 2025;8:395. DOI: [10.1038/s41746-025-01802-z](https://doi.org/10.1038/s41746-025-01802-z).
51. Omar M, Sorin V, Agbareia R, et al. Evaluating and addressing demographic disparities in medical large language models: a systematic review. *Int J Equity Health* 2025;24:57. DOI: [10.1186/s12939-025-02419-0](https://doi.org/10.1186/s12939-025-02419-0).
52. Mello MM, Guha N. Understanding liability risk from using health care artificial intelligence tools. *N Engl J Med* 2024;390:271-278. DOI: [10.1056/NEJMhle2308901](https://doi.org/10.1056/NEJMhle2308901).
53. Duffourc M, Gerke S. Decoding U.S. tort liability in healthcare's black-box AI era: lessons from the European Union. *Stanford Technol Law Rev* 2024;27:1ff.
54. Duffourc M, Gerke S. Generative AI in health care and liability risks for physicians and safety concerns for patients. *JAMA* 2023;330:313-314. DOI: [10.1001/jama.2023.9630](https://doi.org/10.1001/jama.2023.9630).